Introduction to Machine Learning

Evaluation: ROC Basics

ROC Basics

		True Class y		
		+	−	
Pred. $\hat{y}$	+	TP	FP	PPV = $\frac{TP}{TP+FP}$
	−	FN	TN	NPV = $\frac{TN}{FN+TN}$
		TPR = $\frac{TP}{TP+FN}$	TNR = $\frac{TN}{FP+TN}$	Accuracy = $\frac{TP+TN}{TOTAL}$

Learning goals

- Understand why accuracy is not an optimal performance measure for imbalanced labels
- Understand the different measures computable from a confusion matrix
- Be aware that each of these measures has a variety of names

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CLASS IMBALANCE / 2

Classifying all observations as “no disease” (green) yields top accuracy simply because the “disease” occurs so rarely → accuracy paradox.

IMBALANCED COSTS / 2

Imbalanced costs: classifying incorrectly as "no disease" incurs very high cost.

- Problem: if we were able to specify costs precisely, we could evaluate or even optimize on them.
- This important subfield of ML is called **cost-sensitive learning**, which we will not cover in this lecture unit.
- Unfortunately, users find it notoriously hard to come up with precise cost figures in imbalanced scenarios.
- Evaluating "from different perspectives", with multiple metrics, often helps to get a first impression of system quality.

ROC ANALYSIS

- **ROC analysis** is a subfield of ML which studies the evaluation of binary prediction systems.
- ROC stands for "receiver operating characteristics" and was initially developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields – still has the funny name.

http://m.ediciones.org.pk/images/medinlib/f/f9/media39666/large.jpg

LABELS: ROC METRICS

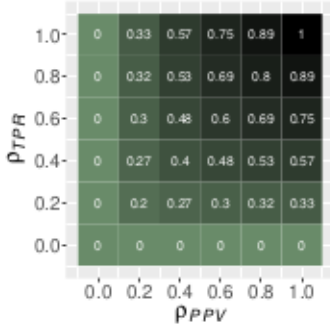
Example:

		Actual Class y		
		Positive	Negative	
$\hat{y}$ **Pred.**	Positive	**True Positive** (TP) = 20	**False Positive** (FP) = 180	Positive predictive value = TP / (TP + FP) = 20 / (20 + 180) = **10%**
	Negative	**False Negative** (FN) = 10	**True Negative** (TN) = 1820	Negative predictive value = TN / (FN + TN) = 1820 / (10 + 1820) ≈ **99.5%**
		True Positive Rate = TP / (TP + FN) = 20 / (20 + 10) ≈ **67%**	True Negative Rate = TN / (FP + TN) = 1820 / (180 + 1820) = **91%**	

https://en.wikipedia.org/wiki/Receiver_operating_characteristic

LABELS: F_1 MEASURE / 2

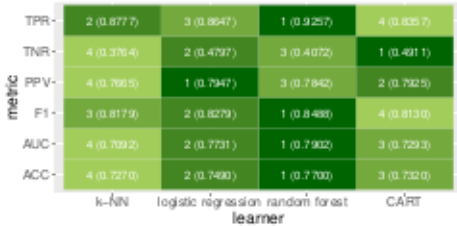
F_1 score for different combinations of ρ_{PPV} & ρ_{TPR}.
$\rightarrow$ Tends more towards the lower of the two combined values.



- A model with $\rho_{TPR} = 0$ (no positive instance predicted as positive) or $\rho_{PPV} = 0$ (no true positives among the predicted) has $\rho_{F_1} = 0$.
- Always predicting "negative": $\rho_{F_1} = 0$.
- Always predicting "positive":
  $\rho_{F_1} = 2 \cdot \rho_{PPV}/(\rho_{PPV} + 1) = 2 \cdot n_+/(n_+ + n)$,
  which will be small when the size of the positive class n_+ is small.

WHICH METRIC TO USE?

- As we have seen, there is a plethora of methods.
  → This leaves practitioners with the question of which to use.
- Consider a small benchmark study.
  - We let k-NN, logistic regression, a classification tree, and a random forest compete on classifying the credit risk data.
  - The data consist of 1000 observations of borrowers' financial situation and their creditworthiness (good/bad) as target.
  - Predicted probabilities are thresholded at 0.5 for the positive class.
  - Depending on the metric we use, learners are ranked differently according to performance (value of respective performance measure in parentheses):



WHICH METRIC TO USE? / 2

- We need not expect overly large discrepancies in general, but neither will we always see an unambiguous picture.
- Different metrics emphasize different aspects of performance.
  $\rightarrow$ The choice should be made in the domain context.
- For practitioners it is vital to understand what should be evaluated exactly, and which measure is appropriate.
  - Regarding credit risk, for instance, defaults are to be avoided, but not at all cost.
  - The bank must undertake a certain risk to remain profitable, so a more balanced measure such as the F_1 score might be in order.
  - On the other hand, a system detecting weapons at an airport should be able to achieve very high true positive rates, even if this comes at the expense of some false alarms.